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ODD END SEMESTER EXAMINATION, DECEMBER-2025

Name of the Course: B.Tech. ECE

Semester: III

Name of the Paper: Advanced Engineering Mathematics

Paper Code: TMA310

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1.	(20 Marks)	
(a)	Show that the function $f(z) = z^n$ satisfies the Cauchy-Riemann equations in polar form. Hence find $f'(z)$.	CO1
(b)	Discuss the analyticity of the function $f(z) = z\bar{z}$ at $z = 0$.	
(c)	Show that $u(x, y) = e^x(\cos y - \sin y)$ is harmonic and find its harmonic conjugate $v(x, y)$. Hence, determine the analytic function $f(z) = u + iv$.	
Q2.	(20 Marks)	
(a)	Evaluate $\oint_C \frac{e^z}{z(z+1)} dz$, where C is the circle $ z = 2$, using the Cauchy residue theorem.	CO2
(b)	Find the Laurent series expansion of $f(z) = \frac{1}{z(1-z)}$ valid in the annulus $0 < z < 1$.	
(c)	Find the bilinear transformation that maps the point $z = 0, 1, \infty$ onto $w = -1, i, 1$ respectively.	
Q3.	(20 Marks)	
(a)	Using Newton Raphson method find a root of $f(x) = x^3 - 4x - 9$ correct to four decimal places	CO3
(b)	Solve $\frac{dy}{dx} = x + y$, with $y(0) = 1$, using Runge-Kutta 4 th order method for $x = 0.4$ by taking $h = 0.2$.	
(c)	For the given data	

	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>4</td></tr><tr><td>f(x)</td><td>1</td><td>3</td><td>7</td><td>63</td></tr></table>	x	0	1	2	4	f(x)	1	3	7	63							
x	0	1	2	4														
f(x)	1	3	7	63														
Estimate $f(3)$ using Lagrange's interpolation formula.																		
Q4.	(20 Marks)																	
(a)	A discrete random variable X has the following distribution: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$P(x)$</td><td>k</td><td>$3k$</td><td>$5k$</td><td>$7k$</td><td>$9k$</td></tr></table> Find k and compute the mean and variance of X.	x	0	1	2	3	4	$P(x)$	k	$3k$	$5k$	$7k$	$9k$	CO4				
x	0	1	2	3	4													
$P(x)$	k	$3k$	$5k$	$7k$	$9k$													
(b)	The average number of telephone calls per minute is 2.5. Find the probability that in a 4 minute interval, exactly 8 calls are received.																	
(c)	The mean height of 500 students is 170 cm with a standard deviation of 5 cm. Find how many students have height between 165 cm and 175 cm. Take $\phi(1) = 0.8413$.																	
Q5.	(20 Marks)																	
(a)	Find the correlation coefficient between X and Y. <table><tr><td>X</td><td>10</td><td>12</td><td>13</td><td>16</td><td>17</td><td>19</td><td>20</td></tr><tr><td>Y</td><td>12</td><td>14</td><td>15</td><td>17</td><td>18</td><td>20</td><td>22</td></tr></table>	X	10	12	13	16	17	19	20	Y	12	14	15	17	18	20	22	CO5
X	10	12	13	16	17	19	20											
Y	12	14	15	17	18	20	22											
(b)	From the following data, obtain the regression equations of Y on X and X on Y, and estimate Y when $X=8$. <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Y</td><td>2</td><td>4</td><td>5</td><td>4</td><td>5</td></tr></table>	X	1	2	3	4	5	Y	2	4	5	4	5					
X	1	2	3	4	5													
Y	2	4	5	4	5													
(c)	Fit a parabola $y = a + bx + cx^2$ to the data: <table><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>Y</td><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td></tr></table>	X	-2	-1	0	1	2	Y	2	1	0	1	2					
X	-2	-1	0	1	2													
Y	2	1	0	1	2													